

Linear Algebra

Friday 2 October 2026 | Main worksheet | Student copy - no solutions

Main route: L1-L6. Course convention: nonzero rows in REF have leading entry 1. Work over the real numbers.

L1 REF, RREF, or neither?

Week 2 tutor activity, p.5. Try alone: 90 seconds; then compare reasons.

For each matrix choose: RREF / REF but not RREF / neither. Circle the leading entries and explain your choice.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

A: _____ because _____

B: _____ because _____

C: _____ because _____

L2 Equations and free variables

Tutor-written activity. Try for 3 minutes. Not the hand-in question.

Write the system as $Ax = b$ and as an augmented matrix. Reduce it and describe every solution.

$$\begin{aligned} x + y + z &= 3 \\ 2x + 2y + 2z &= 6 \end{aligned}$$

Sizes: A is ____ × ____; unknown vector x has ____ entries; b has ____ entries.

Augmented matrix and row operations:

All solutions (include the range of any parameters):

Check: what changes if the right-hand side of the second equation is 7 instead of 6?

L3 has two working pages. Use 8 minutes for your first attempt, then compare and complete it during discussion.

Problem Sheet 2, Q2.2 (additional question). Not Q2.1, the hand-in problem.

$$\begin{array}{rcl} x_1 + x_2 + 2x_3 - 4x_4 & = & -1 \\ -x_1 + x_2 - 2x_3 - 2x_4 & = & -2 \\ -x_1 + 2x_2 - 4x_3 + x_4 & = & -1 \\ 3x_1 - 3x_4 & = & -3 \end{array}$$

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Continue L3. A correct route need not match the tutor's route. Finish by checking the original equations.

L3 Complete, justify, verify

Continuation of Problem Sheet 2, Q2.2. No extra timed task.

Continue your row reduction or write the resulting triangular system:

Back-substitution and solution (x_1, x_2, x_3, x_4) :

Explain why the solution is unique:

Substitute your result into all four original equations:

Reflection: which legal operation did you choose first, and why?

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Check consistency before counting free variables. Optional activities S1-S4 are in the separate supplement.

L4 How many solutions?

Tutor-written activity, related to PS2 Q2.13. Try for 3 minutes.

These are augmented matrices for real variables x, y . State the number of solutions; when consistent, write the whole solution set.

$$\mathbf{U} = \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array} \right]$$

$$\mathbf{N} = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array} \right]$$

$$\mathbf{F} = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array} \right]$$

U: number of solutions / solution set / reason:

N: number of solutions / solution set / reason:

F: number of solutions / solution set / reason:

L5 Challenge the claim

Tutor-written diagnostic. Think of a counterexample for 2 minutes.

“There is a zero row, so there are infinitely many solutions.” Is this always true? Give a counterexample. Explain why legal elementary row operations preserve the solution set.

L6 Exit check - without calculating

Tutor-written check. Write independently for 2 minutes.

1. What extra condition distinguishes RREF from REF?
2. What does a row $[0 \dots 0 \mid 1]$ mean?
3. What must you check before declaring infinitely many solutions?